ENTITY ANALYSIS

CHAPTER-2

2 INTRODUCTION

The Pepperfry E-Commerce Database is designed to efficiently manage all the information and activities involved in an online furniture shopping platform. It consists of multiple entities such as Customer, Product, Category, Brand, Supplier, Cart, Order, Order Details, Payment, Delivery, and Review, each containing attributes that store specific information. These attributes are classified as Simple, Composite, Multivalued, and Derived, while the entities are identified as Strong or Weak based on their dependency. The entities are connected through relationships such as customers placing orders, products belonging to categories, suppliers providing products, and payments being made for orders. These relationships are defined using appropriate cardinality (One-to-One, One-to-Many, and Many-to-Many) and participation constraints (Total and Partial). This well-structured database design ensures data integrity, minimizes redundancy, improves transaction management, and supports the smooth functioning of the Pepperfry e-commerce system.

2.1 Entity: Customer

Attribute	**Key Type**
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	-
Email	Unique Key
Phone_Number	Unique Key
Address	Not Null
Registration_Date	Default (SYSDATE)

2.2 Entity: Product

Attribute	**Key Type**
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Foreign Key

Attribute	Key Type
Brand_ID	Foreign Key
Price	Check (>0)
Stock	Check (>=0)
Material	-
Color	-

2.3 Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

2.4 Entity: Brand

Attribute	Key Type
Brand_ID	Primary Key
Brand_Name	Unique Key
Country	-

2.5 Entity: Supplier

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Contact_Number	Unique Key
Email	Unique Key
Address	-

2.6 Entity: Orders

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	Not Null
Total_Amount	Check (>0)
Order_Status	Not Null
Payment_ID	Foreign Key

2.7 Entity: Order_Details

Attribute	Key Type
Order_Detail_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Check (>0)
Unit_Price	Check (>0)

2.8 Entity: Cart

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Check (>0)

2.9 Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	Not Null
Payment_Status	Not Null
Payment_Date	-

2.10 Entity: Delivery

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	Not Null
Delivery_Date	-
Delivery_Status	Not Null

2.11 Entity: Review

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check (1–5)
Review_Text	-
Review_Date	-

2.12 Entities and Relations

ENTITIES	RELATION	CARDINALITY	TYPE OF PARTICIPATION
Customer – Cart	Owns	One to One	Partial / Total
Customer – Order	Places	One to Many	Partial / Total
Customer – Review	Writes	One to Many	Partial / Total
Customer – Address	Lives At	One to One	Total / Partial
Category – Product	Categorizes	One to Many	Partial / Total
Brand – Product	Manufactures	One to Many	Partial / Total
Supplier – Product	Supplies	One to Many	Partial / Total
Cart – Product	Contains	Many to Many	Partial / Partial
Order – Order_Details	Includes	One to Many	Total / Total
Product – Order_Details	Ordered In	One to Many	Partial / Total
Order – Payment	Paid Through	One to One	Total / Total
Order – Delivery	Delivered By	One to One	Total / Total
Product – Review	Receives	One to Many	Partial / Total

Conclusion

The identification of entities, attributes, keys, and relationships is an important step in designing the Pepperfry E-Commerce Database. The entities were identified based on the major functions of the system, and the required attributes were defined to store accurate and complete information. Primary keys, foreign keys, and other key constraints were used to uniquely identify records and establish connections between entities. The relationships, along with their cardinality and participation constraints, clearly describe how different entities interact with each other. This analysis provides a strong foundation for developing a reliable, well-structured, and efficient database that supports the smooth operation of the Pepperfry e-commerce system.